Field	Details
Date	19 July 2026
Team ID	SB-AIML-2026-0042
Project Name	Emotion Detector - AI-Driven Emotion Detection & Personalized Learning Support P
Maximum Marks	3 Marks

Problem Space

Core Problem

Students in digital learning environments experience a range of emotional states (frustration, confusion, boredom, etc.) while studying. These emotions directly impact their ability to absorb and retain knowledge. However, existing EdTech platforms treat all students identically - delivering the same content regardless of the learner's current emotional or cognitive state.

Pain Points Validated

Pain Point	Validation Source
Students cannot identify why they are stuck	User interviews with engineering students
Generic help resources don't address emotional context	Analysis of existing EdTech platforms
No 24/7 empathetic support available	Survey: 78% of students study after 10 PM with no peer/tutor access
Delayed feedback loop between student struggle and teacher awareness	Educator interviews

Solution Space

Our Solution: Emotion Detector Platform

An AI-powered web application that:

- Accepts student's free-text description of their study problem
- Detects emotional state using BiLSTM + BERT ensemble + rule-based keyword engine
- Generates personalized learning guidance using Google Gemini AI
- Logs and visualizes emotional trends over time

Customer Problem	Solution Feature	Fit Score (1-5)
Can't identify emotional state	Automatic emotion classification (5 classes)	5
Generic help doesn't match mood	Gemini generates emotion-specific	5
No empathetic response	5-section structured guidance with	5
No 24/7 support	Always-online Streamlit Cloud deployment	5
No trend visibility	Analytics dashboard with session history	4
Unsure if approach is correct	Study technique recommendations in guidance	4

Overall Fit Score: 28/30 (93%)

Value Proposition

For	Who	Our product	Is a	That	Unlike	Our solution
Students	who struggle silently while studying	Emotion Detector	AI-powered web app	detects emotions from text and gives tailored guidance	generic search or Q&A forums	responds with empathy and personalized strategy

Assumptions & Risks

Assumption	Risk if Wrong	Mitigation

Assumption	Risk if Wrong	Mitigation
Students will describe problems in English	Lower accuracy for non-English inputs	Add language detection; document English-only constraint
Gemini API will remain on free tier	Cost barrier if API pricing changes	Implement local Gemini-lite fallback
Students willing to type full sentences	Short inputs reduce accuracy	Minimum character validation + example prompts
BiLSTM + BERT ensemble improves accuracy	Ensemble could underperform single model	Compare models; use best performer